HiMCM 2019 Problem A — Summary Sheet

Team GoHiMCM sources: `stage2_baseline_metrics.csv`, `optimal_kiosk_sizing.csv`, `thunderstorm_stress_comparison.csv`, `autonomous_planning_summary.json`

Problem. Size terminal charging kiosks under CapEx $\leq$ \$80,000 and area $\leq 40\,\mathrm{m}^2$. Each traveler is a BDI passenger agent with SOC and boarding slack. Agents choose a fast kiosk, a slow locker, a power bank, or abort.

Model. SimPy maps three resource pools: 65 W fast kiosk, 15 W locker, power-bank bank. Arrivals use a 16-hour NHPP with three peaks. LangGraph + local Ollama (`qwen2.5:7b`) propose packs. FastMCP tools call the DES and check budget. Acceptance: mean wait ≤ 5 min; reneging probability ≤ 0.05.

Stage-2 baseline (12/15/2, normal day). $n = 441$ arrivals; 110 anxious seekers. Mean wait = 0.00 min. Reneging probability = 0.00. Fast utilization = 5.72%; slow utilization = 24.24%. Max system load = 17.

Pareto sizing (rush densify; reneging $\leq 5\%$). 216 unique packs; 7 non-dominated points. Knee under the \$80,000 line: pack **8/16/0**. CapEx = **\$35,200** (44% of CapEx cap). Area = $38.4\,\mathrm{m}^2$. Rush mean wait = 0.0115 min; reneging probability = 0.00. Cheapest Pareto pack is 4/8/0 (wait 5.91 min). Knee wait is 0.0115 min. Mean wait falls by 99.8% between those packs.

Thunderstorm stress (17:00**; 40% delayed 2–4 h; SOC drain $\times$2; $n = 621$).**

KPI	Baseline 12/15/2	Knee 8/16/0
CapEx (USD)	49,000	35,200
Mean wait (min)	4.591	2.347
Reneging probability	0.0229	0.0114
Max wait-queue length	6	6

Mean wait falls by 48.9%. Reneging probability falls by 50.3%. CapEx falls by 28.2% versus the empirical pack.

Recommendation. Install pack 8/16/0 first. Keep CapEx headroom for gate-cluster add-ons. Offer a free first 15 minutes on the fast kiosk to raise turnover. Re-run multi-seed storm tests before capital release.